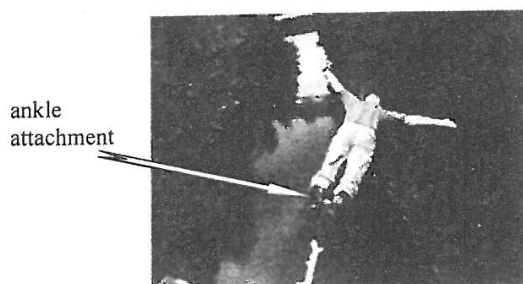


6. Read the following description about '**Bungee jumping**' and answer the questions that follow.

Bungee jumping is an activity that involves jumping from a tall structure while the person is connected to it via a thick elastic cord. When the bungee jumper jumps, the cord stretches after falling a certain distance. The bungee jumper is momentarily at rest at the lowest point but then bounces back up into the air. The bungee jumper continues to oscillate up and down a few times before he comes to a complete stop.



A simple 'ankle attachment' (as shown in the above photo) can be used to secure the jumper to the cord. However, due to accidents where the ankle attachment became detached from the bungee jumper, many operators now use a 'full body harness'.



When answering the following questions, neglect the effects of air resistance.

- (a) (i) Describe the acceleration of the bungee jumper during the first downward fall to the lowest point. (3 marks)

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Answers written in the margins will not be marked.

- (a) (ii) State the energy change during the period from the beginning of the jump to the moment when the bungee jumper is at the lowest point of his first downward fall. (2 marks)

- (b) In terms of the net force acting on the bungee jumper, explain why the cord has to be elastic. (2 marks)

- (c) In terms of contact area, explain why a 'full body harness' is less likely to cause injuries to or detach from the bungee jumper than a simple 'ankle attachment' during a fall. (2 marks)

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7.

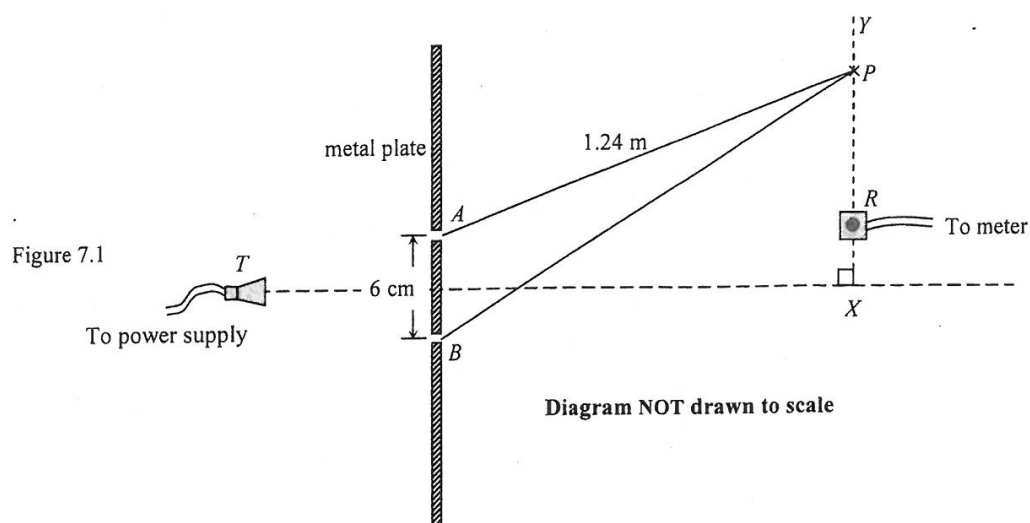


Figure 7.1 shows a set-up for the study of interference of microwaves. Microwaves of wavelength 2 cm emitted from a transmitter T pass through two slits A and B formed by metal plates. The slits are separated by 6 cm as shown. A probe R connecting to a meter is moved from X to Y to detect the intensity of microwaves received. Transmitter T and point X are equidistant from A and B .

- (a) Calculate the frequency of the microwaves. (2 marks)

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Answers written in the margins will not be marked.

- (b) (i) The meter shows alternate maxima and minima when R moves along XY . Explain. (2 marks)

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9. An ideal gas is at temperature 0°C and pressure $1.01 \times 10^5 \text{ N m}^{-2}$.

(a) *(i) Find the volume of one mole of the gas.

(2 marks)

(ii) Find the number of molecules in 1 m^3 of the gas under the conditions given.

(2 marks)

*(b) At a certain location above the Earth in space where the pressure is $2.75 \times 10^{-10} \text{ N m}^{-2}$, 1 m^3 of the gas contains about 1.00×10^{11} molecules. Assuming the gas is ideal, estimate the temperature at the location (in K).

(2 marks)

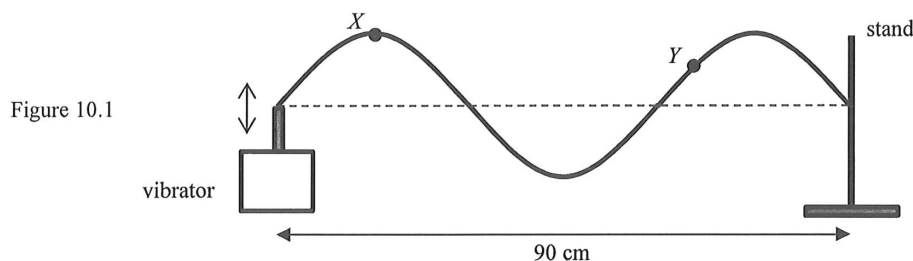
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10. (a) A student uses the set-up in Figure 10.1 to investigate stationary waves on a stretched string. When the frequency of the vibrator is set at 100 Hz, a stationary wave is formed between the vibrator and the stand, which are 90 cm apart.



Particles X and Y on the string are at maximum displacement from their respective equilibrium positions at the moment shown in Figure 10.1.

- Label the positions of all antinodes (using 'A') and nodes (using 'N') on Figure 10.1. (2 marks)
- Compare the oscillations of X and Y in terms of their amplitude and phase. (2 marks)

Answers written in the margins will not be marked.

- Sketch on Figure 10.1 the shape of the string just after 0.0075 s. (1 mark)
- If the frequency of the vibrator is increased gradually, at what frequency will the next stationary wave pattern be observed? Show your working. (2 marks)

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(ii) Estimate the magnitude of the acceleration of the sphere.

(1 mark)

(iii) What force provides the centripetal force for the sphere ?

(1 mark)

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Answers written in the margins will not be marked.